SG No. 227 O_{h}^{7} $Fd\bar{3}m$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `-43m`

No.	position	mapping
1	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]`
2	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]`

Table 2: Wyckoff site: `8b`, site symmetry: `-43m`

No.	position	mapping
1	$\left[\frac{3}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]`
2	$\left[\frac{1}{8},\,\frac{5}{8},\,\frac{1}{8}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]`

Table 3: Wyckoff site: `16c`, site symmetry: `.-3m`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,5,9,14,19,24,25,29,33,38,43,48]`
2	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{1}{2}\right]$	`[2,7,12,13,17,21,26,31,36,37,41,45]`
3	$\left[\frac{1}{4},\,\frac{1}{2},\,\frac{3}{4}\right]$	`[3,8,10,15,20,22,27,32,34,39,44,46]`
4	$\left[\frac{1}{2},\,\frac{3}{4},\,\frac{1}{4}\right]$	`[4,6,11,16,18,23,28,30,35,40,42,47]`

Table 4: Wyckoff site: `16d`, site symmetry: `.-3m`

No.	position	mapping
1	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[1,5,9,14,19,24,25,29,33,38,43,48]`
2	$\left[\frac{1}{4},\,\frac{3}{4},\,0\right]$	`[2,7,12,13,17,21,26,31,36,37,41,45]`
3	$\left[\frac{3}{4},\,0,\,\frac{1}{4}\right]$	`[3,8,10,15,20,22,27,32,34,39,44,46]`
4	$\left[0,\,\frac{1}{4},\,\frac{3}{4}\right]$	`[4,6,11,16,18,23,28,30,35,40,42,47]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3m`

No.	position	mapping
1	$[x,\,x,\,x]$	`[1,5,9,38,43,48]`
2	$\left[\frac{3}{4}-x,\,\frac{1}{4}-x,\,x+\frac{1}{2}\right]$	`[2,7,12,37,41,45]`
3	$\left[\frac{1}{4}-x,\,x+\frac{1}{2},\,\frac{3}{4}-x\right]$	`[3,8,10,39,44,46]`
4	$\left[x+\frac{1}{2},\,\frac{3}{4}-x,\,\frac{1}{4}-x\right]$	`[4,6,11,40,42,47]`
5	$\left[x+\frac{3}{4},\,x+\frac{1}{4},\,\frac{1}{2}-x\right]$	`[13,17,21,26,31,36]`
6	$[-x,\,-x,\,-x]$	`[14,19,24,25,29,33]`
7	$\left[x+\frac{1}{4},\,\frac{1}{2}-x,\,x+\frac{3}{4}\right]$	`[15,20,22,27,32,34]`
8	$\left[\frac{1}{2}-x,\,x+\frac{3}{4},\,x+\frac{1}{4}\right]$	`[16,18,23,28,30,35]`

Table 6: Wyckoff site: `48f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[x,\ \frac{1}{8},\ \frac{1}{8}\right]$	`[1,4,42,43]`
2	$\left[\frac{3}{4}-x,\ \frac{1}{8},\ \frac{5}{8}\right]$	`[2,3,41,44]`
3	$\left[\frac{1}{8},\ x,\ \frac{1}{8}\right]$	`[5,8,38,39]`
4	$\left[\frac{5}{8},\ \frac{3}{4}-x,\ \frac{1}{8}\right]$	`[6,7,37,40]`
5	$\left[\frac{1}{8},\ \frac{1}{8},\ x\right]$	`[9,12,45,48]`
6	$\left[\frac{1}{8},\ \frac{5}{8},\ \frac{3}{4}-x\right]$	`[10,11,46,47]`
7	$\left[\frac{7}{8},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	`[13,16,30,31]`
8	$\left[\frac{7}{8},\ -x,\ \frac{7}{8}\right]$	`[14,15,29,32]`
9	$\left[x+\frac{3}{4},\ \frac{3}{8},\ \frac{3}{8}\right]$	`[17,20,26,27]`
10	$\left[\frac{1}{2}-x,\ \frac{7}{8},\ \frac{3}{8}\right]$	`[18,19,25,28]`
11	$\left[\frac{7}{8},\ \frac{3}{8},\ \frac{1}{2}-x\right]$	`[21,24,33,36]`
12	$\left[\frac{3}{8},\ \frac{3}{8},\ x+\frac{3}{4}\right]$	`[22,23,34,35]`

Table 7: Wyckoff site: `96g`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,38]`
2	$\left[\frac{3}{4}-x,\ \frac{1}{4}-x,\ z+\frac{1}{2}\right]$	`[2,37]`
3	$\left[\frac{1}{4}-x,\ x+\frac{1}{2},\ \frac{3}{4}-z\right]$	`[3,39]`
4	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	`[4,40]`
5	$[z,\ x,\ x]$	`[5,48]`
6	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$	`[6,47]`
7	$\left[\frac{3}{4}-z,\ \frac{1}{4}-x,\ x+\frac{1}{2}\right]$	`[7,45]`
8	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-x\right]$	`[8,46]`
9	$[x,\ z,\ x]$	`[9,43]`
10	$\left[\frac{1}{4}-x,\ z+\frac{1}{2},\ \frac{3}{4}-x\right]$	`[10,44]`
11	$\left[x+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	`[11,42]`
12	$\left[\frac{3}{4}-x,\ \frac{1}{4}-z,\ x+\frac{1}{2}\right]$	`[12,41]`
13	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-z\right]$	`[13,26]`
14	$[-x,\ -x,\ -z]$	`[14,25]`
15	$\left[x+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	`[15,27]`
16	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	`[16,28]`
17	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-x\right]$	`[17,36]`
18	$\left[\frac{1}{2}-x,\ z+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[18,35]`
19	$[-x,\ -z,\ -x]$	`[19,33]`
20	$\left[x+\frac{1}{4},\ \frac{1}{2}-z,\ x+\frac{3}{4}\right]$	`[20,34]`
21	$\left[z+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-x\right]$	`[21,31]`
22	$\left[z+\frac{1}{4},\ \frac{1}{2}-x,\ x+\frac{3}{4}\right]$	`[22,32]`
23	$\left[\frac{1}{2}-z,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[23,30]`
24	$[-z,\ -x,\ -x]$	`[24,29]`

Table 8: Wyckoff site: `96h`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ y,\ -y]$	`[1,19]`
2	$\left[\frac{3}{4},\ \frac{1}{4}-y,\ \frac{1}{2}-y\right]$	`[2,17]`
3	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y+\frac{3}{4}\right]$	`[3,20]`
4	$\left[\frac{1}{2},\ \frac{3}{4}-y,\ y+\frac{1}{4}\right]$	`[4,18]`
5	$[-y,\ 0,\ y]$	`[5,14]`
6	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ \frac{1}{4}-y\right]$	`[6,16]`
7	$\left[y+\frac{3}{4},\ \frac{1}{4},\ y+\frac{1}{2}\right]$	`[7,13]`
8	$\left[y+\frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}-y\right]$	`[8,15]`
9	$[y,\ -y,\ 0]$	`[9,24]`
10	$\left[\frac{1}{4}-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[10,22]`
11	$\left[y+\frac{1}{2},\ y+\frac{3}{4},\ \frac{1}{4}\right]$	`[11,23]`
12	$\left[\frac{3}{4}-y,\ y+\frac{1}{4},\ \frac{1}{2}\right]$	`[12,21]`
13	$[0,\ -y,\ y]$	`[25,43]`
14	$\left[\frac{1}{4},\ y+\frac{3}{4},\ y+\frac{1}{2}\right]$	`[26,41]`
15	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{4}-y\right]$	`[27,44]`
16	$\left[\frac{1}{2},\ y+\frac{1}{4},\ \frac{3}{4}-y\right]$	`[28,42]`
17	$[y,\ 0,\ -y]$	`[29,38]`
18	$\left[y+\frac{1}{2},\ \frac{1}{4},\ y+\frac{3}{4}\right]$	`[30,40]`
19	$\left[\frac{1}{4}-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	`[31,37]`
20	$\left[\frac{3}{4}-y,\ \frac{1}{2},\ y+\frac{1}{4}\right]$	`[32,39]`
21	$[-y,\ y,\ 0]$	`[33,48]`
22	$\left[y+\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[34,46]`
23	$\left[\frac{1}{2}-y,\ \frac{1}{4}-y,\ \frac{3}{4}\right]$	`[35,47]`
24	$\left[y+\frac{1}{4},\ \frac{3}{4}-y,\ \frac{1}{2}\right]$	`[36,45]`

Table 9: Wyckoff site: `192i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{3}{4}-x,\ \frac{1}{4}-y,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[\frac{1}{4}-x,\ y+\frac{1}{2},\ \frac{3}{4}-z\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{4}-z\right]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-y\right]$	`[6]`
7	$\left[\frac{3}{4}-z,\ \frac{1}{4}-x,\ y+\frac{1}{2}\right]$	`[7]`
8	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-y\right]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$\left[\frac{1}{4}-y,\ z+\frac{1}{2},\ \frac{3}{4}-x\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	`[11]`
12	$\left[\frac{3}{4}-y,\ \frac{1}{4}-z,\ x+\frac{1}{2}\right]$	`[12]`
13	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-z\right]$	`[13]`
14	$[-y,\ -x,\ -z]$	`[14]`
15	$\left[y+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	`[16]`
17	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-y\right]$	`[17]`

continued ...

Table 9

No.	position	mapping
18	$\left[\frac{1}{2}-x,\, z+\frac{3}{4},\, y+\frac{1}{4}\right]$	[18]
19	$[-x,\, -z,\, -y]$	[19]
20	$\left[x+\frac{1}{4},\, \frac{1}{2}-z,\, y+\frac{3}{4}\right]$	[20]
21	$\left[z+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-x\right]$	[21]
22	$\left[z+\frac{1}{4},\, \frac{1}{2}-y,\, x+\frac{3}{4}\right]$	[22]
23	$\left[\frac{1}{2}-z,\, y+\frac{3}{4},\, x+\frac{1}{4}\right]$	[23]
24	$[-z,\, -y,\, -x]$	[24]
25	$[-x,\, -y,\, -z]$	[25]
26	$\left[x+\frac{1}{4},\, y+\frac{3}{4},\, \frac{1}{2}-z\right]$	[26]
27	$\left[x+\frac{3}{4},\, \frac{1}{2}-y,\, z+\frac{1}{4}\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{4},\, z+\frac{3}{4}\right]$	[28]
29	$[-z,\, -x,\, -y]$	[29]
30	$\left[\frac{1}{2}-z,\, x+\frac{1}{4},\, y+\frac{3}{4}\right]$	[30]
31	$\left[z+\frac{1}{4},\, x+\frac{3}{4},\, \frac{1}{2}-y\right]$	[31]
32	$\left[z+\frac{3}{4},\, \frac{1}{2}-x,\, y+\frac{1}{4}\right]$	[32]
33	$[-y,\, -z,\, -x]$	[33]
34	$\left[y+\frac{3}{4},\, \frac{1}{2}-z,\, x+\frac{1}{4}\right]$	[34]
35	$\left[\frac{1}{2}-y,\, z+\frac{1}{4},\, x+\frac{3}{4}\right]$	[35]
36	$\left[y+\frac{1}{4},\, z+\frac{3}{4},\, \frac{1}{2}-x\right]$	[36]
37	$\left[\frac{1}{4}-y,\, \frac{3}{4}-x,\, z+\frac{1}{2}\right]$	[37]
38	$[y,\, x,\, z]$	[38]
39	$\left[\frac{3}{4}-y,\, x+\frac{1}{2},\, \frac{1}{4}-z\right]$	[39]
40	$\left[y+\frac{1}{2},\, \frac{1}{4}-x,\, \frac{3}{4}-z\right]$	[40]
41	$\left[\frac{1}{4}-x,\, \frac{3}{4}-z,\, y+\frac{1}{2}\right]$	[41]
42	$\left[x+\frac{1}{2},\, \frac{1}{4}-z,\, \frac{3}{4}-y\right]$	[42]
43	$[x,\, z,\, y]$	[43]
44	$\left[\frac{3}{4}-x,\, z+\frac{1}{2},\, \frac{1}{4}-y\right]$	[44]
45	$\left[\frac{1}{4}-z,\, \frac{3}{4}-y,\, x+\frac{1}{2}\right]$	[45]
46	$\left[\frac{3}{4}-z,\, y+\frac{1}{2},\, \frac{1}{4}-x\right]$	[46]
47	$\left[z+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-x\right]$	[47]
48	$[z,\, y,\, x]$	[48]